CHURN PREDICTION COMPETITION REPORT

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1. Introduction and Problem Setup

The objective of this project is to develop a machine learning model capable of predicting whether a user will churn within a short future time horizon, based on historical interaction data. Specifically, the goal is to identify users who are likely to churn in the **10 days** following the end of the observation period, **which concludes on November 20, 2018**

A user is considered to have churned if they visit the page *"Cancellation Confirmation"* during the prediction window. The target variable is therefore binary, taking value 1 for users who churn and 0 otherwise. The problem can thus be framed as a **binary classification problem**, where the objective is to estimate churn probabilities for users in the test set.

The dataset consists of **event-level observations**, where each row corresponds to a user interaction at a given point in time. A central challenge of the project lies in transforming raw event logs into informative user-level features while **strictly respecting the temporal structure** of the data. In addition, churn events represent only a small fraction of the observations, leading to a pronounced **class imbalance** that directly influences model choice and evaluation metrics.

The project proceeds as follows. First, the dataset is analyzed and transformed into temporally consistent user-level snapshots. Next, several machine learning models are tested and compared in order to identify the approach that best captures the patterns associated with future churn. Finally, results are presented and discussed.

2. Data Description and Label Construction

2.1 Dataset Structure

The dataset contains approximately **17.5 million event-level observations** and **19 variables**, including categorical, numerical, textual, and datetime features. No duplicate rows are present. Core variables include a unique user identifier (userId), event timestamps, session-related information, and categorical attributes describing the type of interaction performed.

Some variables, such as length, song, and artist, exhibit missing values, while most categorical features (e.g. page, gender, level) are fully observed. The temporal coverage spans from **early October 2018** to **November 20, 2018**, which coincides with the end of the observation window. Initial preprocessing steps convert timestamps into datetime format and remove redundant or non-informative fields.

2.2 Temporal Framing and Snapshot Construction

To account for the dynamic nature of user behavior, a **rolling snapshot approach** is adopted. Instead of relying on a single train–test split, multiple snapshots are constructed over time using automatically generated cutoff dates.

Cutoffs start after a minimum history requirement of 14 days and progress in 7-day increments until the last cutoff allowing for a full 10-day prediction horizon. For each cutoff date, features are computed using all events observed up to and including the cutoff, while labels are derived exclusively from events

occurring in the subsequent prediction window. This design enforces a strict temporal separation between inputs and targets.



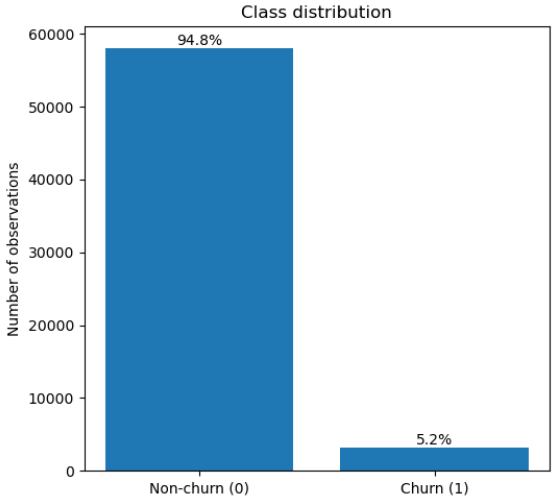
The rolling-window approach allows each user to contribute multiple observations over time, provided sufficient historical activity is available. The final dataset includes **17,129 unique users**, with a median of **four snapshots per user**, increasing the effective sample size while preserving temporal consistency.

2.3 Churn Definition and Target Variables

The primary target variable (churn) is defined as a binary indicator equal to 1 if a user visits the page *"Cancellation Confirmation"* within **10 days after a given cutoff date**, and 0 otherwise.

To ensure a correct temporal separation between features and labels, churn is determined solely using events that occur after the cutoff date. Users who have already churned before a cutoff are excluded, as they are no longer at risk.

In addition to the short-term churn indicator, a secondary target variable (churn_ever) is constructed to indicate whether a user churns at any point after the cutoff date. Two separate models are trained on these targets, and final predictions are obtained by averaging the corresponding predicted probabilities.



The resulting dataset is highly imbalanced, with approximately **5.2%** of observations labeled as churn within the 10-day horizon. This characteristic motivates the use of evaluation metrics that appropriately account for class imbalance.

2.4 Feature Construction Overview

Feature engineering is performed by aggregating event-level information into **user-level features** for each snapshot, using only data available prior to the cutoff date.

The constructed features can be grouped into several categories. First, **activity-related features** summarize the overall intensity of user interactions, such as the total number of actions, total number of song-related events, and total active time measured across sessions. Second, **behavioral indicators** capture specific types of interactions, including errors and negative feedback actions.

To capture **short-term dynamics**, additional features are computed over recent time windows of 3, 7, and 14 days prior to the cutoff, measuring the volume of events and selected interaction types in each window. **Rates** and normalized indicators are also derived to account for differences in user activity levels.

Further features describe **temporal aspects** of user behavior, such as the number of days since the last observed event, user lifetime measured from registration time, and simple trend indicators comparing recent and past activity. Finally, basic demographic and technical attributes are included **through binary indicators and one-hot encoding**, such as gender, geographical state, and operating system inferred from the user agent string.

After feature construction, non-predictive identifiers and raw categorical fields are removed, resulting in a clean user-level dataset suitable for supervised learning.

3. Modeling and Evaluation Strategy

3.1 Preliminary Modeling

In an initial phase, churn prediction was approached as a static classification problem using a stratified random train–test split. Logistic Regression, Random Forest, and XGBoost models were evaluated, with hyperparameters tuned via GridSearchCV and balanced accuracy used as the primary optimization metric.

As shown in Table 1, XGBoost consistently outperformed both the linear baseline and the bagging-based model across all evaluation metrics, achieving the **highest balanced accuracy**, F1-score, and ROC AUC. Feature selection techniques were also explored at this stage but were discarded due to a systematic degradation in performance.

These results motivated the exclusive use of XGBoost in the subsequent, temporally consistent modeling framework.

Table 1

Model	Balanced Accuracy	F1-score	ROC AUC
Logistic Regression	0.713	0.538	0.775
Random Forest	0.687	0.512	0.786
XGBoost	**0.724**	**0.558**	**0.799**

3.2 Temporal Modeling and Data Splitting

In the final framework, churn prediction explicitly focuses on future user behavior. A time-based outer split is applied, assigning the most recent **20% of cutoff dates** to the test set and using earlier snapshots for training. This strategy ensures that the test set strictly represents future periods with respect to the training data.

Within the training set, hyperparameter tuning is performed using **GroupKFold cross-validation**, grouping observations by user identifier to prevent leakage across folds. Numerical features are winsorized at the 1st and 99th percentiles based on training data only, while feature scaling is handled within the modeling pipeline using a robust scaler.

3.3 Final Model and Probability Blending

Based on the results of the preliminary modeling phase, **XGBoost** is selected as the preferred model for the temporal framework, due to its ability to capture non-linear patterns and interaction effects among user-level behavioral features while remaining robust under class imbalance.

Two separate XGBoost classifiers are trained for churn and churn_ever, respectively. Class imbalance is addressed through **class-weighting** via the scale_pos_weight parameter, computed from the training data for each target.

Hyperparameters are optimized using **Optuna** and, for each trial, model performance is evaluated using the mean balanced accuracy obtained from GroupKFold cross-validation on the training set. The hyperparameter search space includes tree depth, learning rate, number of estimators, subsampling ratios, and regularization-related parameters.

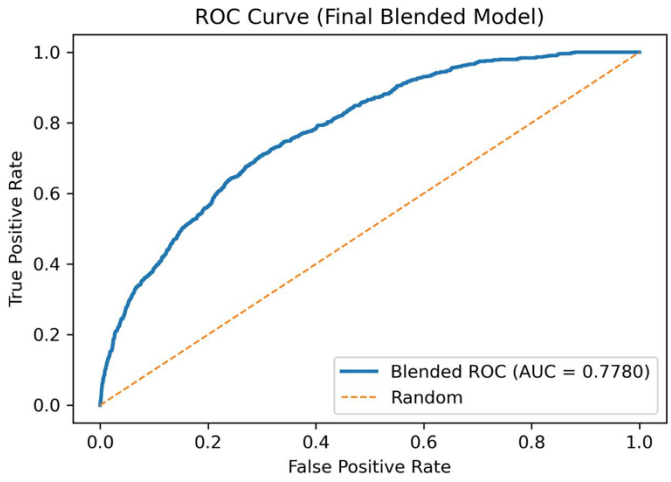
Finally, the predicted probabilities produced by the two models are **linearly combined**, with the blending weight selected through cross-validation using balanced accuracy as the optimization criterion.

4. Results and Conclusion

4.1 Final Results

When evaluated on the held-out test set, the final blended model achieves the following performance:

- **Balanced Accuracy:** 0.7195
- **ROC AUC:** 0.8072
- **F1-score:** 0.2330



These results indicate strong discriminatory performance despite the severe class imbalance characterizing the dataset. In particular, the high ROC AUC highlights the model's effectiveness in ranking users by churn risk, while balanced accuracy confirms robust performance across classes.

The relatively low F1-score is consistent with the rarity of churn events and reflects the trade-off between recall and precision in highly imbalanced classification problems. For this reason, balanced accuracy and ROC AUC provide a more informative assessment of model performance.

4.2 Conclusion

This project presents a temporally and logically consistent framework for churn prediction. By aligning feature construction, validation strategy, and evaluation metrics with the forward-looking nature of the problem, the proposed approach effectively addresses the competition objective. While performance is constrained by class imbalance and the short prediction horizon, the results underscore the importance of temporal awareness and appropriate evaluation in churn modeling.